The hosting model for the previous ASP.NET framework was a relatively complex one, relying on Windows IIS to provide the web-server hosting. In a cross-platform environment, this kind of symbiotic relationship isn't possible, so an alternative hosting model has been adopted—one that separates web applications from the underlying host. This opportunity has led to the development of Kestrel, a fast, cross-platform HTTP server on which ASP.NET Core can run.

Instead of the previous design, whereby IIS calls into specific points of your application, ASP.NET Core applications are console applications that self-host a web server and handle requests directly, as shown in figure 2.6. This hosting model is conceptually much simpler and allows you to test and debug your applications from the command line, though it doesn't necessarily remove the need to run IIS (or the equivalent) in production.

ASP.NET Core and reverse proxies

You can expose ASP.NET Core applications directly to the internet so that Kestrel receives requests directly from the network. That approach is fully supported. It's more common, however, to use a reverse proxy between the raw network and your application. In Windows, the reverse-proxy server typically is IIS; in Linux or macOS, it might be NGINX, HAProxy, or Apache. There's even an ASP.NET Core-based reverse proxy library called YARP (https://microsoft.github.io/reverse-proxy) that you can use to build your own reverse proxy.

A reverse proxy is software responsible for receiving requests and forwarding them to the appropriate web server. The reverse proxy is exposed directly to the internet, whereas the underlying web server is exposed only to the proxy. This setup has several benefits, primarily security and performance for the web servers.

You may think that having a reverse proxy *and* a web server is somewhat redundant. Why not have one or the other? Well, one benefit is the decoupling of your application from the underlying operating system. The same ASP.NET Core web server, Kestrel, can be cross-platform and used behind a variety of proxies without putting any constraints on a particular